

Allocation Process

exec() -> PGD allocated (0x00100000)

main() -> Text page fault: PT for Text, Data, and BSS (0x00101000) & Text page (0x00102000)

d[3]=b[0] -> Data page fault: Data page (0x00103000)

BSS page (0x00104000)

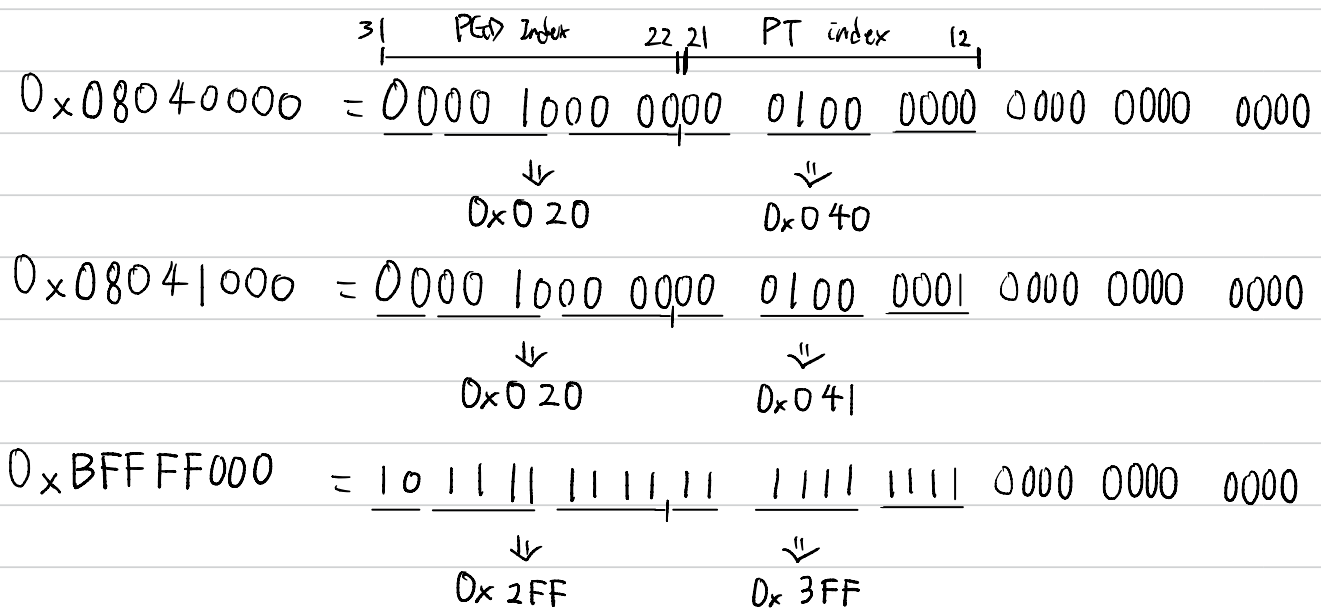
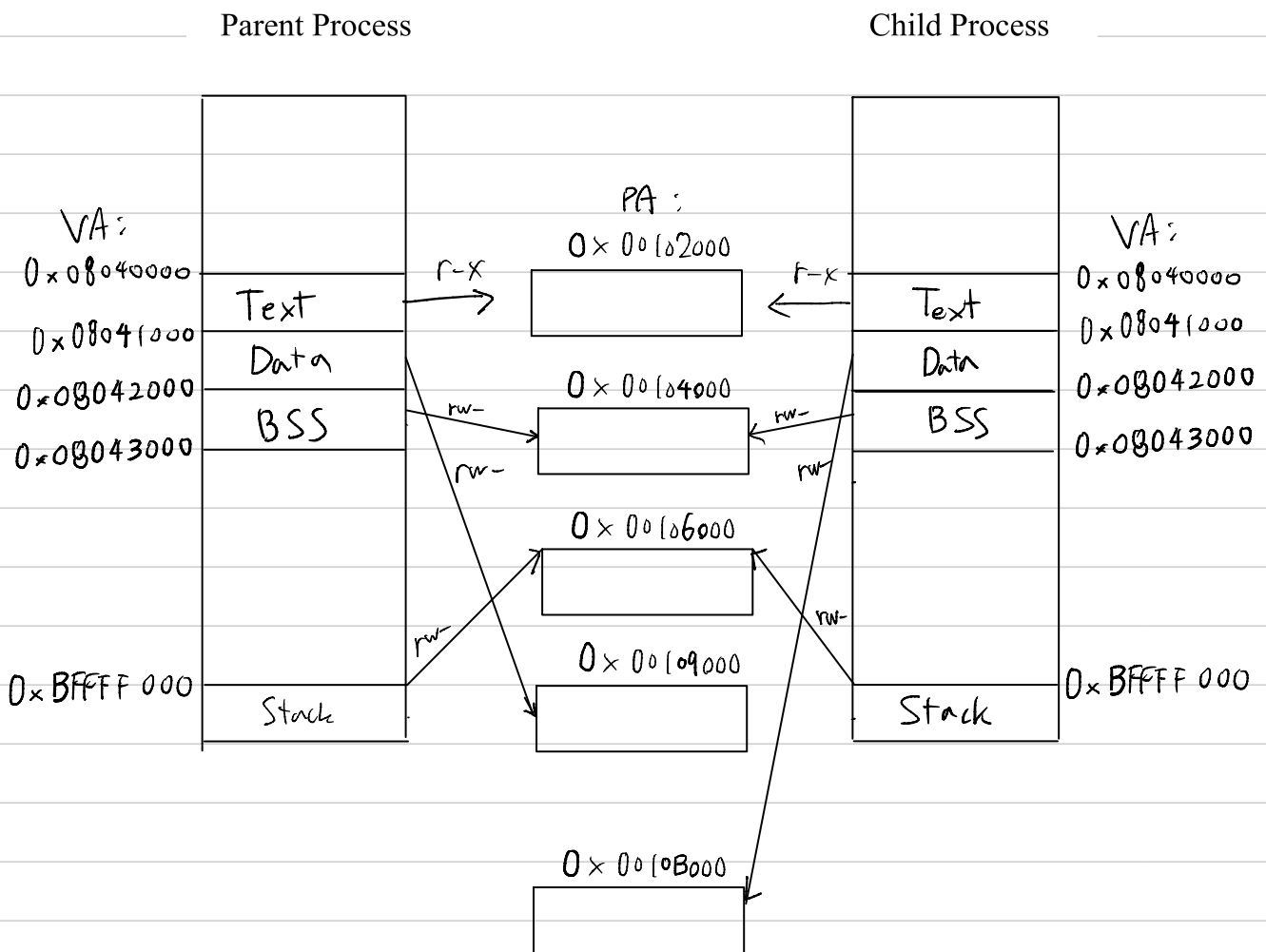
int pid -> Stack page fault: PT for Stack(0x00105000) & Stack page (0x00106000)

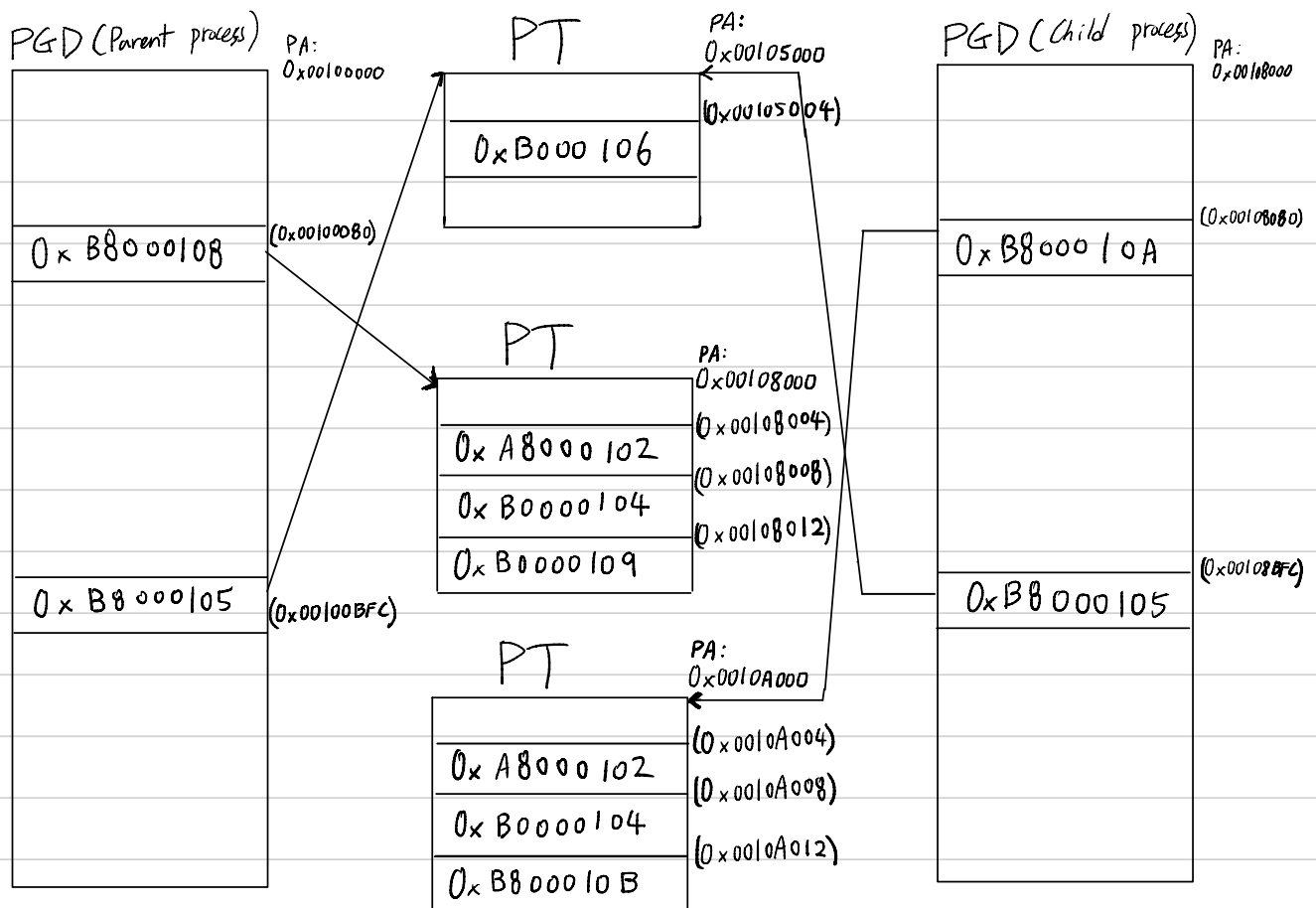
fork() -> BOTH parent and child trigger COW: child PGD (0x00107000)

Whatever triggered first: COW -> PT (0x00108000) & Data page (0x00109000)

Whatever triggered next: COW -> PT (0x0010A000) & Data page (0x0010B000)

* I assumed that parent process was triggered earlier than child process.





PURWXDA N

0xA8 = 10101000

0xB0 = 10110000

0xB8 = 10111000

ex) B8000107

Permission
Bits

Determined by PFN of PT at 0x00107000

ex) $(0x00100080) = \text{PGD base} + \underbrace{\text{index}}_{0x020} \times 4 \text{ bytes}$

$(0x00100BFC) = \text{PGD base} + \underbrace{\text{index}}_{0x2FF} \times 4 \text{ bytes}$